Problem 5: Fairness Transfer across States and Years

ICT 5143: AFGML Capstone

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07/11/26

Research question and decision context

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- **Question:** with only 200 or 1,000 target labels, does updating ≤ 2 Bayesian-network factors keep equalized odds better than global adaptation?
- **Falsified if** the Holm-adjusted paired bootstrap shows no gain, or balanced accuracy drops by more than 0.02.

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- Closest idea: Schrouff et al. (2022) *diagnose* transfer failures with a graph, but do not repair them.
- **BN-LFA (ours):** update only the ≤ 2 shifted factors that matter for the decision, then refit the fair thresholds.

Data and evaluation population

- ACS PUMS 1-Year, CA, TX, NY $\times$ 2018, 2019: six domains (Folktables check passed).

	CA18	TX18	NY18	CA19	TX19	NY19
persons	195,665	135,924	103,021	197,193	138,293	104,330
groups $\geq$100 pos./neg. (test)	10	8	7	10	8	8

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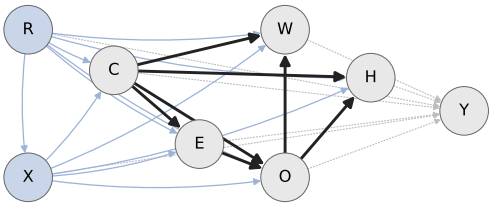
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- Small groups (American Indian, Alaska Native, Pacific Islander) are reported, but only as exploratory.

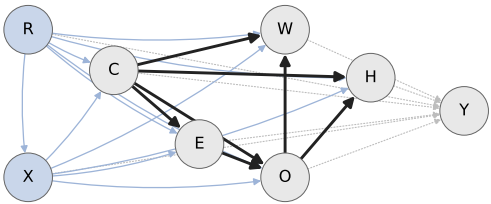
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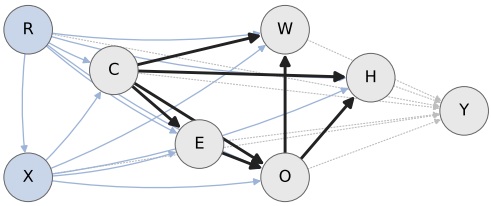
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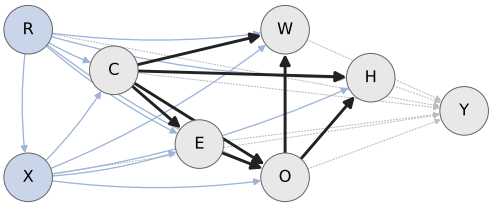
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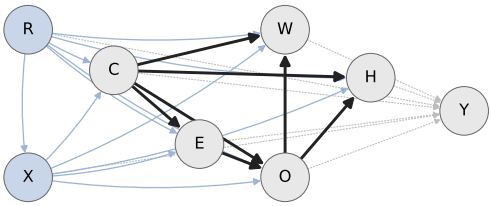
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- Query $P(G, \mathrm{pa}(Y), Y)$ gives each group's TPR and FPR.
- VE $=$ clique tree $=$ enumeration $=$ pgmpy (error 1.3×10^{-15}).

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- Uses race and sex at prediction time (features and group thresholds).

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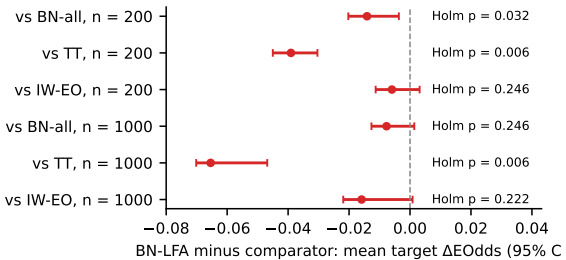
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- Paired household bootstrap ($B = 2000$); Holm over 6 confirmatory tests.
- Plus a synthetic simulation, ablations and robustness tests.

Main result and uncertainty

Method, $n = 1000$	BA	Δ_{EOdds}
EO-S (source rule)	0.771	0.150
TT	0.772	0.205
IW-EO (closest)	0.769	0.156
BN-all	0.770	0.147
BN-LFA	0.770	**0.140**

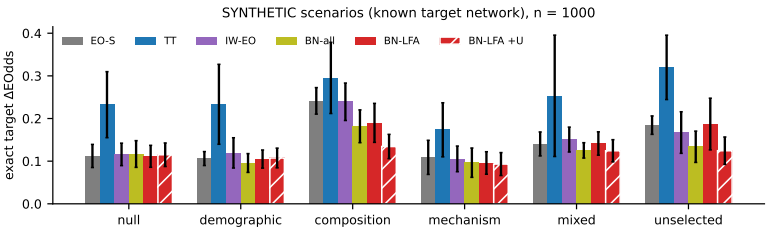
Mean of 9 transfers $\times$ 5 seeds; lower Δ_{EOdds} is fairer.



Answer: partially supported. BN-LFA is fairer than TT (both budgets) and BN-all ($n = 200$), but not than IW-EO or the unchanged source rule.

Ablation, robustness and failure case

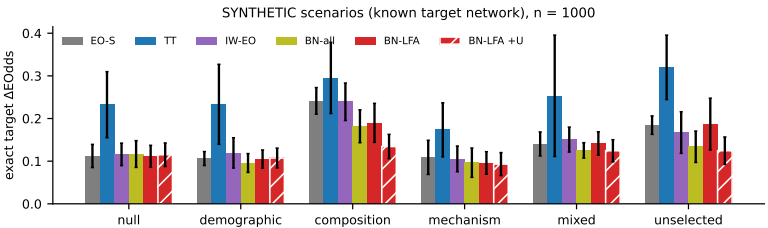
- **Ablation** ($n = 1000$): random factors 0.152 and all factors 0.147, vs BN-LFA 0.140.



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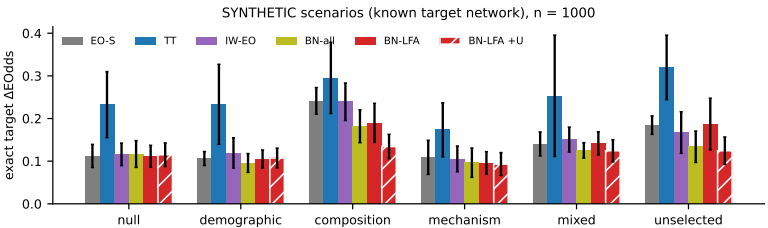
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- **Mechanism:** the diagnostic predicts the fairness change (Spearman 0.88).
- **Failure:** three shifted factors but only two allowed: no better than EO-S (0.187 vs 0.184).



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- **Limits:** associational graph; 3 states, 2 years; small groups; the \$50k label; race and sex used at decision time.
- **Reproduce:** `github.com/aslammugutkhan/fairness-transfer` (commit `2eaef71`): *Run all* in the notebook on Colab (13 min).

Backup: mandatory evidence and references

Requirement (Problem 5)	Evidence in `results/`
E1–E3: within-domain, transfers, budgets	`exp_transfer_summary.csv`, Figs. 2–3
E4: composition vs mechanism simulation	`sim_all_runs.csv`, Fig. 7
E5: method comparison	`confirmatory_tests_holm.csv`
Ablations; robustness tests	`ablation_d_eodds.csv`, `robustness_summary.csv`

Full matrix: `docs/requirements_matrix.md`. Key references: Ding et al. 2021; Shao et al. 2024; Hardt et al. 2016; Schrouff et al. 2022; Singh et al. 2021; Pearl & Bareinboim 2011 (16 papers in `docs/related_work.md`).

Backup: inference and statistical validation

- CPTs normalised; factor algebra, evidence and missing values checked (notebook self-checks).
- 20 random queries: all exact methods agree to 1.3×10^{-15}. Elimination order changes cost only.
- Every local Markov statement is a d-separation; largest residual dependence 0.047 bits.
- Model-based group TPRs within sampling noise for 89 % of groups.
- Bootstrap resamples households; the same resamples for all methods (paired).

Backup: demonstration and individual defence

- Run notebook section 2.12 (a few seconds).
- $P(Y{=}1 \mid \text{Black, female, graduate}) = 0.6636$ by VE, clique tree and enumeration.
- CA18→TX18 with 200 labels: BN-LFA updates E and R; Δ_{EOdds} 0.143 vs 0.156 (EO-S), 0.237 (TT).
- Why R barely matters: equalized odds conditions on the group, so race mix alone does not move error rates.
- Saved trace: `demo/demo_trace.txt`.